

GPS versus Local Knowledge Classroom Run Kit

TOK Cognitive Science Activity Bank • Knowledge and Technology • 25 min

Category	Knowledge and Technology	Time	25 min
Difficulty	Easy	ID	gps-versus-local-knowledge

Big idea: Data can be precise and incomplete.

One-Minute Teacher Brief

Students compare precise digital knowledge with situated local knowledge.

Student Prompt

Inspect GPS versus Local Knowledge Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

GPS versus Local Knowledge Stimulus A	Student-facing stimulus 1: A route-planning scenario where the map suggests a faster but impractical route. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
GPS versus Local Knowledge Stimulus B	Student-facing stimulus 2: Local knowledge cards: school gate closed, roadworks, unsafe crossing, steep hill. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
GPS versus Local Knowledge Reveal / Twist	Student-facing stimulus 3: A decision table balancing speed, safety, reliability, and lived experience. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — Data can be precise and incomplete.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting GPS versus Local Knowledge Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from GPS versus Local Knowledge Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	When is local knowledge better than technological knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Knowledge and Technology.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how data can be precise and incomplete.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: When is local knowledge better than technological knowledge?

Setup Checklist

- Prepare a starter stimulus using: A route-planning scenario where the map suggests a faster but impractical route.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

- A route-planning scenario where the map suggests a faster but impractical route.
- Local knowledge cards: school gate closed, roadworks, unsafe crossing, steep hill.
- A decision table balancing speed, safety, reliability, and lived experience.

Resources To Use

- Resource pack: <resources/gps-versus-local-knowledge/index.html>
- Card deck CSV: <resources/gps-versus-local-knowledge/cards.csv>
- Visual prompt: <resources/gps-versus-local-knowledge/visual-prompt.svg>
- Printable student worksheet with observation, inference, evidence, and confidence columns.
- Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.
- Classroom run kit with prompt cards, example stimuli, and debrief moves.
- PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Give students a route recommended by an app.
2. Add local knowledge: roadworks, locked gate, dangerous crossing, staircase, informal shortcut, crowding at certain times.
3. Groups design the best route for different users.
4. Compare algorithmic optimisation with human judgement.

Debrief Moves

- Knowledge question: When is local knowledge better than technological knowledge?
- Does data capture everything relevant?
- Can technology make us less attentive knowers?

Teacher Quick Script

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...